Operationalising predictive dynamics of song and speech with information-theoretic models

Felix Haiduk[*1], Nicholas Harley[2], John McBride[3], and Tudor Popescu[4]
[*]Corresponding Author: haiduk.felix@gmail.com
[1]Acoustics Research Institute, Austrian Academy of Sciences, Vienna, Austria
[2]Vrije Universiteit Brussel, Brussel, Belgium
[3]University of Vienna, Vienna, Austria
[4]Faculty of Psychology, University of Vienna, Vienna, Austria

We suggest that one driver for language and music evolution was the employment of different coordination goals - choric and dialogic vocalisations, leading to differences in mutual predictability of interlocutors. Here, we investigated whether vocalisation features from a cross-cultural dataset representing a music-speech continuum differ in predictability. We hypothesised music to be more predictable than speech and found initial evidence but also more complex patterns by employing different information-theoretical analyses. We discuss the results with regard to current evolutionary theorising.

1. **Background**

Coordination of vocalisations is an important task for a variety of species, for example in duetting or chorusing behaviour (Ravignani et al., 2014). In humans, spoken language requires the coordination of turn-taking, while sung music often requires the coordination of multiple concurrent sound productions. Such coordination tasks rely on predictive processing for humans (Levinson & Torreira, 2015) and at least some animals (e.g. Benichov et al., 2016). Haiduk & Fitch (2022) therefore proposed that certain vocalisation features such as discrete pitch in music are adapted to afford coordination goals such as choric (concurrent) or dialogic (turn-taking) production, explaining vocal differences between song and speech. Since current evolutionary theorising emphasises the influence of the behavioural niche on evolution, such constraints on signal structure may have influenced the evolution of music and language.

2. Aims

In the current study, we took a step back and investigated whether cross-cultural acoustical feature differences prototypical of song or spoken language indicated predictability differences. Because music affords choric coordination with several concurrent sound streams, we hypothesised that sung music would be more predictable than spoken language across time and pitch dimensions.

3. Methods

We assessed predictability in song, instrumental music, and recited and descriptive speech from a cross-cultural dataset, comprising examples from 21 language families/cultural backgrounds from all continents (Ozaki et al., 2024). We quantified predictability using both 4-gram analysis and the information-theoretic model IDyOMS (Pearce, 2005). The former forms a baseline emphasising the role of local statistics. The latter is a variable-order Markov model operationalising auditory perception through "viewpoints" such as pitch and IOI. It supports multiple interacting viewpoints and long-range contexts, capturing prediction based on higher-order structure. Using both models allowed us to distinguish predictability based on local vs higher-order cues. To this end, we computed information content (IC) and entropy (H).

4. Results

Preliminary results show that n-gram entropy decreased from descriptive to recited speech to song and instrumental music, forming a continuum with the lowest uncertainty in instrumental music. However, we also observed more complex patterns of IC and H differed depending on the model (n-gram/IDyOMS) and the number and choice of viewpoints, suggesting predictive structure not captured by 4-gram analysis.

5. Discussion and conclusion

We discuss the results regarding their robustness as well as current limitations and future directions of operationalising predictive dynamics using information-theoretic measures to investigate the evolution of song and speech. We conclude that the influence of coordination goals, vocal features, predictive processes, and cognitive trait evolution might have important implications for our understanding of music and language evolution.

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